Week 2: Matrix multiplication and rank

Linear Algebra I

Bowen Gan

Fall 2026

These notes were drafted with the help of ChatGPT from my handwritten notes and the discussion used to prepare the recitation.

1. Returning to a square system

Last time we compared different solutions of $Ax = b$. If x_0 is one solution, every other solution has the form $x_0 + v$, where $Av = 0$. The homogeneous equation describes all the changes we can make to a solution without changing its right-hand side.

For an $n \times n$ matrix A, this led to four equivalent statements:

(a) $Ax = 0$ has only the zero solution.
(b) $Ax = b$ has exactly one solution for at least one $b \in \mathbb{R}^n$.
(c) $Ax = b$ has exactly one solution for every $b \in \mathbb{R}^n$.
(d) $\operatorname{rank} A = n$.

Here rank has its course definition: the number of nonzero rows in an echelon form. To recall the role of the square-matrix assumption, if there are no nonzero homogeneous solutions, there are no free variables. All n columns must be pivot columns. Since there are also exactly n rows, the reduced row echelon form is I_n, and every right-hand side gives a unique solution.

The converse implications are short. If a particular right-hand side has a unique solution x_0, then $x_0 + v$ can be a solution only when $v = 0$. Thus the homogeneous equation has no nonzero solutions. If every right-hand side has a unique solution, we can in particular take $b = 0$.

We can now add another equivalent statement: A is invertible. If A^{-1} exists, the unique solution is $x = A^{-1}b$. Conversely, suppose every system $Ax = b$ has a unique solution. Solve $Ax_j = e_j$ for each standard coordinate vector e_j, and set $X = [\, x_1 \ \cdots \ x_n \,]$. Then $AX = I$. Moreover,

$$A(XA - I) = (AX)A - A = 0.$$

Each column of $XA - I$ is a homogeneous solution, so $XA = I$ as well. Hence $X = A^{-1}$. Invertibility and the existence of A^{-1} are two ways to say the same thing.

If determinant notation is familiar, $\det A \neq 0$ is another equivalent condition. We will not need determinants in the arguments below. The Moore–Penrose inverse extends the idea of an inverse to singular and rectangular matrices, but it does not supply a two-sided inverse when one does not exist. That is a separate topic.

2. Multiplication as composition

A matrix does more than store the coefficients of an equation. An $m \times n$ matrix A defines a map $T_A : \mathbb{R}^n \to \mathbb{R}^m$ by $T_A(x) = Ax$. It is linear: $A(x + y) = Ax + Ay$ and $A(cx) = cAx$. These two properties say that applying the map respects addition and scalar multiplication.

Suppose B takes an input x to $y = Bx$, and A then takes y to $z = Ay$. Substituting the first expression into the second gives

$$z_i = \sum_k a_{ik} y_k = \sum_k a_{ik} \sum_j b_{kj} x_j = \sum_j \left(\sum_k a_{ik} b_{kj} \right) x_j.$$

The coefficient of x_j in the ith output is exactly $(AB)_{ij}$. Thus the rule for matrix multiplication expresses composition: $T_{AB} = T_A \circ T_B$. The factor on the right acts first.

Rotation and reflection make the order visible. Write

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The first rotates the plane counterclockwise through θ; the second reflects it across the x-axis. For $e_1 = (1,0)^T$,

$$R_\theta S e_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad S R_\theta e_1 = \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix}.$$

When $\sin\theta \neq 0$, the two outputs differ, so $R_\theta S \neq S R_\theta$. At $\theta = \pi/2$, for instance, one order sends e_1 upward and the other sends it downward. Rotation and reflection commute only in the exceptional cases $\theta \in \pi\mathbb{Z}$.

Composition is associative. Applying C, then B, then A produces the same result whether we first group BC or AB. This explains $(AB)C = A(BC)$. Associativity changes the grouping, not the order of the operations. Some particular operations do commute: two rotations about the origin in the plane satisfy

$$R_\theta R_\varphi = R_{\theta+\varphi} = R_\varphi R_\theta.$$

This is a property of these rotations, not of all matrix products.

There are two further differences from real-number arithmetic. Nonzero matrices can have zero product. For example,

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad \text{satisfy} \qquad HK = 0.$$

The map K keeps only the vertical component, and H then removes that component. Both maps are nonzero, but their composition is zero.

Likewise, $AB = AC$ and $A \neq 0$ do not imply $B = C$: take $A = H$, $B = K$, and $C = 0$. An invertible left factor can be cancelled, because multiplying by A^{-1} gives $B = C$. More precisely, for a square matrix A, cancellation in *every* equality $AB = AC$ is equivalent to invertibility. If A is singular, choose $v \neq 0$ with $Av = 0$; the nonzero matrix whose first column is v and whose other columns are zero has the same product with A as the zero matrix.

3. Rank through one example

The number of pivots is easy to compute, but it is useful to know what that number measures. Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + x_3 \\ x_2 + x_3 \\ x_1 + x_2 + 2x_3 \end{pmatrix}.$$

If we put $u = x_1 + x_3$ and $v = x_2 + x_3$, the output is $(u, v, u+v)^T$. Conversely, every vector of this form is obtained by taking the input $(u, v, 0)^T$. Thus the set of all outputs is exactly the plane $w = u + v$.

This set is called the *image* of A, written

$$\operatorname{im} A = \{Ax : x \in \mathbb{R}^3\} = \{(u, v, w)^T : w = u + v\}.$$

For a general $m \times n$ matrix with columns $a_1, \ldots, a_n$, the image consists of all linear combinations of the columns, since

$$Ax = x_1 a_1 + \cdots + x_n a_n$$

The image is therefore also called the *column space*.

We need a little vocabulary to describe the size of such a set. A nonempty subset of $\mathbb{R}^m$ closed under linear combinations is a *linear subspace*. The span of a list of vectors is the set of all their linear combinations. A list $v_1, \ldots, v_r$ is *linearly independent* if $c_1 v_1 + \cdots + c_r v_r = 0$ forces all the coefficients to vanish. An independent list that spans a subspace is a *basis*; its length is the *dimension* of that subspace.

Why does the length not depend on the basis? If a space is spanned by r vectors, any $r+1$ vectors in it are dependent. Express those $r+1$ vectors using the r generators. A relation between them can be found by solving r homogeneous equations in $r+1$ unknown coefficients, which has a nonzero solution. Comparing two bases in both directions shows that their lengths agree. For a finite spanning list, a basis can be found by keeping a vector exactly when it is not in the span of those already kept.

In our example,

$$(u, v, u+v)^T = u(1,0,1)^T + v(0,1,1)^T.$$

The two displayed vectors are independent, as their first two coordinates show. They form a basis of $\operatorname{im} A$, so its dimension is 2. Row reduction also gives two pivots. The general equality $\operatorname{rank} A = \dim \operatorname{im} A$ will follow from the row-rank and column-rank argument below.

The inputs sent to zero form another subspace, called the *kernel*:

$$\ker A = \{x : Ax = 0\} = \{t(-1,-1,1)^T : t \in \mathbb{R}\}.$$

It is the homogeneous solution set, also called the nullspace in the first week's notes. Here the equations are $x_1 + x_3 = 0$ and $x_2 + x_3 = 0$, so taking $x_3 = t$ gives the displayed expression. Increasing the input by a multiple of $(-1,-1,1)^T$ leaves the output unchanged.

For $b = (u, v, w)^T$, the equation $Ax = b$ is consistent exactly when $w = u + v$. When it is consistent, all its solutions are

$$x = \begin{pmatrix} u \\ v \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

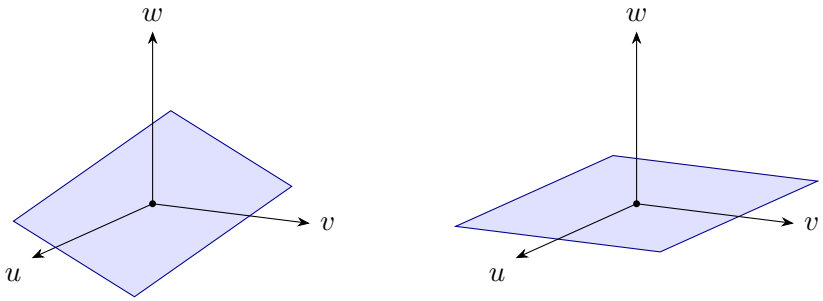
Thus the same matrix has two independent output directions and one free direction in its homogeneous solution set. In this example, $\dim \ker A + \operatorname{rank} A = 1 + 2 = 3$. This is the simplest instance of the rank–nullity formula, which we will return to after the rank calculation.

It is helpful to keep the two ambient spaces distinct. For an $m \times n$ matrix, the kernel is a subspace of the input space $\mathbb{R}^n$, while the image is a subspace of the output space $\mathbb{R}^m$. Every vector in the kernel maps to the *zero vector* in the output space; the whole input space maps onto the image.

Now subtract the first two rows from the third, $R_3 \leftarrow R_3 - R_1 - R_2$:

$$A \longrightarrow U = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The homogeneous equations have not changed, so $\ker U = \ker A$. The image has changed: $Ux = (x_1 + x_3, x_2 + x_3, 0)^T$, and consequently $\operatorname{im} U$ is the plane $w = 0$. The two images have the same dimension, but they are different planes. This distinction is the reason to examine row operations more carefully.



$\operatorname{im} A:\ w = u + v$ $\qquad$ $\operatorname{im} U:\ w = 0$

Figure 1: Row operations change the image plane while preserving its dimension. The shaded patches represent parts of planes extending in all directions.

4. What row operations preserve

Suppose B is obtained from an $m \times n$ matrix A by elementary row operations. Each operation is left multiplication by an invertible elementary matrix. If the operations occur in the order $E_1, \ldots, E_k$, then

$$B = E_k \cdots E_1 A = PA, \qquad P = E_k \cdots E_1.$$

The later operations appear farther to the left. The matrix P is invertible because every step can be undone.

There are two different statements about rows and columns. The *row space*, consisting of all linear combinations of the rows, stays the same. Indeed, $B = PA$ expresses each row of B as a combination of rows of A, and $A = P^{-1}B$ gives the reverse inclusion. Thus $\operatorname{Row}(B) = \operatorname{Row}(A)$.

For the columns, the vectors themselves change, but their relations keep exactly the same coefficients. If $A = [\, a_1 \ \cdots \ a_n \,]$ and $B = [\, b_1 \ \cdots \ b_n \,]$, then $b_j = Pa_j$. Hence

$$\sum_{j=1}^{n} c_j b_j = 0 \quad \Longleftrightarrow \quad P\left(\sum_{j=1}^{n} c_j a_j\right) = 0 \quad \Longleftrightarrow \quad \sum_{j=1}^{n} c_j a_j = 0.$$

The last equivalence uses invertibility. In particular, any chosen columns are independent before the operations exactly when the corresponding columns are independent afterwards. This preservation of coefficients applies to *column* relations; rows retain their span, but need not retain their original relations with the same coefficients.

Writing the coefficients as a vector x shows that $Bx = 0$ if and only if $Ax = 0$. Thus the kernel is unchanged. The image transforms according to

$$\operatorname{im} B = \{PAx : x \in \mathbb{R}^n\} = P(\operatorname{im} A).$$

Here $P(\operatorname{im} A)$ means that we apply P to every vector in the old image. In the preceding example, $P(u, v, w)^T = (u, v, w - u - v)^T$, which takes the plane $w = u + v$ to the plane $w = 0$.

An invertible matrix preserves the dimension of every subspace it transforms. To see this directly, let $v_1, \ldots, v_r$ be a basis of a subspace W. Their images $Pv_1, \ldots, Pv_r$ span $P(W)$ by linearity. If $\sum_i c_i P v_i = 0$, multiplying by P^{-1} gives $\sum_i c_i v_i = 0$, so all the coefficients vanish. The images therefore form a basis of the same length. This proves $\dim P(W) = \dim W$.

The kernel equality concerns homogeneous equations. For a nonhomogeneous equation the right-hand side must change too:

$$Ax = b \quad \Longleftrightarrow \quad PAx = Pb.$$

This is why Gaussian elimination applies row operations to the entire augmented matrix.

The language of dimension extends beyond finite lists of coordinates. A physical example is the space of smooth displacement profiles $f : [0, L] \to \mathbb{R}$ of an ideal string fixed at its ends, so $f(0) = f(L) = 0$. Profiles can be added and multiplied by real numbers. The functions $\sin(k\pi x/L)$, for positive integers k, give arbitrarily large independent families: multiplying a finite linear relation by one of these sine functions and integrating from 0 to L isolates its coefficient. Thus this space has no finite basis. This is an example from classical vibrations; the familiar finite-dimensional spaces are enough for our matrix arguments.

5. Changing both sides of a matrix

The same reasoning also describes column operations. More generally, let $P \in \mathrm{GL}_m(\mathbb{R})$ and $Q \in \mathrm{GL}_n(\mathbb{R})$, where $\mathrm{GL}_d(\mathbb{R})$ denotes the invertible $d \times d$ matrices, and put $B = PAQ$. Then

$$\operatorname{im} B = P(\operatorname{im} A), \qquad \ker B = Q^{-1}(\ker A).$$

For the image, Qx ranges over all of $\mathbb{R}^n$ as x does, so $\{PAQx : x \in \mathbb{R}^n\} = \{PAu : u \in \mathbb{R}^n\}$. For the kernel,

$$PAQx = 0 \quad \Longleftrightarrow \quad AQx = 0 \quad \Longleftrightarrow \quad Qx \in \ker A.$$

This is exactly the assertion $x \in Q^{-1}(\ker A)$. The two image dimensions and the two kernel dimensions are consequently unchanged. Once we identify rank with image dimension below, we also obtain $\operatorname{rank}(PAQ) = \operatorname{rank} A$.

These formulas distinguish two effects. Left multiplication applies an invertible transformation to the outputs; right multiplication replaces the input x by Qx. For a square matrix, the special expression $S^{-1}AS$ represents the same linear map after changing the basis in both its domain and codomain in the same way. The columns of S are the new basis vectors in old coordinates. In this case the coordinate descriptions of both the image and the kernel are transformed by S^{-1}.

6. Row rank, column rank, and normal form

We can now compare three descriptions of rank. The row rank is the dimension of the row space, or equivalently the largest number of independent rows. The column rank is the dimension of the column space, or equivalently the largest number of independent columns. Since the column space is the image, column rank is $\dim \operatorname{im} A$.

For either rows or columns, one can select a maximal independent sublist from the finite list of vectors. Maximal means that no further vector from the list can be added without creating a dependence. Such a sublist spans every vector in the list: otherwise the missing vector could be added. It is therefore a basis, and all such selections have the same length by the dimension argument in Section 3. No choice principle is needed for this finite selection.

The row and column ranks are both unchanged by elementary row operations. The row-space equality proves this for rows, and the preservation of column relations proves it for columns. These facts were proved separately, without assuming that the two ranks agree.

Let U be the reduced row echelon form of A, with r nonzero rows and pivot columns $j_1, \ldots, j_r$. The reduced form is unique, although an ordinary echelon form need not be. To compare the ranks, it is enough to inspect this reduced matrix.

Its r nonzero rows are independent. In a relation $\lambda_1\rho_1 + \cdots + \lambda_r\rho_r = 0$, the entry in pivot column j_i is precisely λ_i, so every coefficient is zero. Since all other rows are zero, the row rank is r.

The pivot columns are $e_1, \ldots, e_r$ in $\mathbb{R}^m$, and are independent. Every other column has zero entries below row r, so it is a combination of these pivot columns. Hence the column rank is also r. For the zero matrix both ranks are zero. We have proved, for every matrix,

$$\operatorname{rowrank} A = \operatorname{colrank} A = \operatorname{rank} A = \dim \operatorname{im} A.$$

Thus counting pivots, counting independent rows, and counting independent output directions all give the same number.

For instance, in the reduced matrix

$$U = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

the two nonzero rows are independent, the first two columns are independent, and the third column is their sum. The rank is 2 from either viewpoint. To find a basis of the *original* column space, take the columns of A at these pivot positions, not the columns of U: column relations are preserved, but the image space may have moved.

We can also finish the rank–nullity argument. Suppose A has n columns and rank r. The equation $Ux = 0$ has $n - r$ free variables. Set one free variable to 1 and the others to 0, and solve for the pivot variables; do this once for each free variable. The resulting $n - r$ solutions are independent because their free coordinates are the standard coordinate vectors. They span every homogeneous solution because the free coordinates determine the pivot coordinates uniquely. Since $\ker A = \ker U$, they give a basis of $\ker A$, and therefore

$$\dim \ker A + \operatorname{rank} A = n.$$

The right-hand side is the number of input coordinates. No assumption that A is square is needed.

The reduced form can be simplified further if we also allow column operations. First move its pivot columns to the front. The resulting matrix has block form

$$\begin{pmatrix} I_r & F \\ 0 & 0 \end{pmatrix}.$$

Subtract suitable combinations of the first r columns from each remaining column to remove F. Equivalently,

$$\begin{pmatrix} I_r & F \\ 0 & 0 \end{pmatrix} \begin{pmatrix} I_r & -F \\ 0 & I_{n-r} \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

The right-hand factor is invertible, with F in place of $-F$ in its inverse. All row and column operations used are invertible. Thus, for every $m \times n$ matrix A of rank r, there exist $L \in \mathrm{GL}_m(\mathbb{R})$ and $C \in \mathrm{GL}_n(\mathbb{R})$ such that

$$LAC = J_r, \qquad J_r = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \in M_{m,n}(\mathbb{R}).$$

Blocks of size zero are omitted when $r = m$ or $r = n$. This is the *rank normal form*.

For our running example, row operations already gave U. Subtracting the first two columns from the third, $C_3 \leftarrow C_3 - C_1 - C_2$, then gives

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

In the resulting input and output coordinates, the map is simply $(u, v, t)^T \mapsto (u, v, 0)^T$. In general, J_r keeps the first r coordinates and sends the remaining input directions to zero. This explains why rank is the complete invariant for matrices of a fixed size under invertible row and column operations: every rank-r matrix reduces to the same J_r.

The two coordinate changes here are independent. This is more flexible than a similarity transformation $S^{-1}AS$, which uses the same new basis on both sides. For example, $2I$ remains $2I$ under every similarity transformation, although invertible row operations take it to I.

7. Two homework arguments

The first homework problem returns to the square-matrix equivalences from the beginning. Exercise 2.4/9 asks us to prove that if $I + AB$ is invertible, then $I + BA$ is invertible, where A and B are square matrices of the same size. There is no assumption that A or B is invertible.

We can prove invertibility by showing that a homogeneous equation has only the zero solution. Suppose $(I + BA)x = 0$. We want to bring the known invertible matrix $I + AB$ into the argument. Multiplying by A does exactly that:

$$0 = A(I + BA)x = (I + AB)Ax.$$

Since $I+AB$ is invertible, $Ax = 0$. Substituting this into the original equation gives $x+BAx = x = 0$. Therefore $I + BA$ is invertible. The identity $A(I + BA) = (I + AB)A$ comes from associativity; it does not exchange A and B.

Exercise 2.4/2(2) asks why column operations can be used to compute an inverse. Suppose A is invertible and apply the same sequence of elementary column operations to A and to I. If the corresponding elementary matrices are $E_1, \ldots, E_k$, in the order of application, set $Q = E_1 \cdots E_k$. The two results are

$$A \longmapsto AQ, \qquad I \longmapsto IQ = Q.$$

If the operations reduce A to I, then $AQ = I$, so $Q = A^{-1}$. The identity matrix has therefore become the inverse.

For row operations, we usually place the matrices side by side. For column operations, placing them one above the other makes a single operation act on both:

$$[\, A \mid I \,] \xrightarrow{\text{row operations}} [\, I \mid A^{-1} \,], \qquad \begin{pmatrix} A \\ I \end{pmatrix} \xrightarrow{\text{column operations}} \begin{pmatrix} I \\ A^{-1} \end{pmatrix}.$$

Both algorithms record the invertible operations used to turn A into I. The difference is the side on which those operations multiply.